

Pharmacovigilance Text Analytics

Real World Problem

After drugs are released to market, pharmaceutical companies and regulatory bodies (like Health Canada, FDA) are obligated to monitor the safety of drugs after approval, called “pharmacovigilance”. This critical function involves analyzing vast quantities of unstructured text from medical case reports, scientific literature, and patient-generated content to identify potential Adverse Drug Events (ADEs).

Currently, identifying these events requires manual review by medical experts, which is expensive, slow and difficult. As the scale of the medical data grows, this becomes impossible to monitor manually. Failure to detect ADEs early can increase patient risk and prompt regulatory action. A system that can automatically flag whether a body of text is associated with an ADE meets both a business and clinical need.

High Level Analytics Approach

Our primary goal is to build a reliable text classification system. We will develop and evaluate a model to determine if a given piece of text (at the sentence level) describes an ADE.

- Text preprocessing: tokenization, lemmatization and part-of-speech tagging.
- Information Extraction: Apply Named Entity Recognition (NER) to isolate drug names and adverse effect mentions from text, potentially standardizing using ICD-10.
- Classification Modeling: Develop and compare multiple classification models including baseline model (e.g., Naïve Bayes, Nearest Neighbor) to predict ADE relevance at the sentence level.
- Evaluation: model will be evaluated based on its accuracy and ability to minimize false negatives to ensure potential ADEs are not missed.

Data Source and Extraction Plan

Data source: https://huggingface.co/datasets/SetFit/ade_corpus_v2_classification

The dataset consists of thousands of sentences labeled as either an adverse drug event or not.

Extension Opportunity: Extract case reports of drugs from PubMed API. Feed sentences into model to flag those with a high probability of describing an ADE. Cross reference against FDA FAERS API to see if adverse drug events spiked for that drug at the same time case report was published.